

• Homework 2

Consider a spacecraft composed of a main cylindrical body B_1 and four appendages (attached to B_1)

Body B_1 , with radius R and length L

$$m_{B_1} = 2000 \text{ Kg}$$

$$R = 1 \text{ m}$$

$$L = 4 \text{ m}$$

Bodies $B_2 - B_5$, with radius r_A and length l

$$m_{B_2} = m_{B_3} = m_{B_4} = m_{B_5} = 100 \text{ Kg}$$

$$r_A = 0.05 \text{ m}$$

$$l = \begin{matrix} 2 \text{ m} & \text{or} & 6 \text{ m} \\ (1) & & (2) \end{matrix}$$

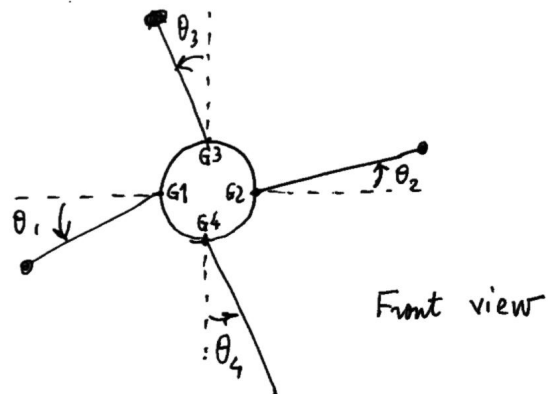
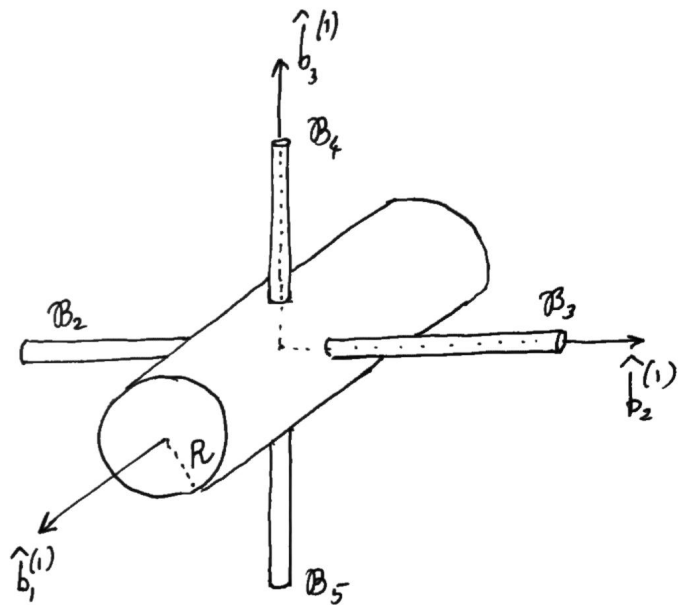
Each joint is associated with an elastic torque and a friction torque

$$\underline{T_i} = (-K_1 \theta_i - K_2 \dot{\theta}_i) \hat{b}_1^{(1)} \text{ torque on body } i \text{ (} i = 2, 3, 4, 5 \text{)}$$

$$\underline{T_1} = - \sum_{i=2}^5 \underline{T_i}$$

where

$$\begin{cases} K_1 = 100 \text{ Nm} & \text{for the elastic torque} \\ K_2 = 50 \text{ Nm sec} & \text{for the friction torque} \end{cases}$$



Front view

$G1 \div G4$ are revolute joints
(1 degree of freedom rotations)

The spacecraft travels a circular orbit of radius $R_0 = 7000 \text{ km}$ and the inertial frame coincides with $\underline{B}_1$ at t_0 . The position and velocity of the center of mass of B_1 at t_0 are

$$\underline{\vec{v}}_{p1}(t_0) = \begin{bmatrix} 0 & \sqrt{\frac{\mu}{R_0}} & 0 \end{bmatrix} \underline{N}^T$$

$$\underline{N}^T = \underline{B}_1^T(t_0)$$

$$\underline{\vec{r}}_{p1}(t_0) = \begin{bmatrix} R_0 & 0 & 0 \end{bmatrix} \underline{N}^T$$

The remaining initial conditions are

$$\underline{\vec{\omega}}_1(t_0) = \begin{bmatrix} 1 & 0.1 & 0.1 \end{bmatrix} \text{sec}^{-1} \underline{B}_1^T$$

$$\theta_1(t_0) = -30 \text{ deg} \quad \theta_2(t_0) = 30 \text{ deg} \quad \theta_3(t_0) = -30 \text{ deg} \quad \theta_4(t_0) = 10 \text{ deg}$$

$$\sigma_1(t_0) = -0.2 \text{ sec}^{-1} \quad \sigma_2(t_0) = 0.2 \text{ sec}^{-1} \quad \sigma_3(t_0) = 0.1 \text{ sec}^{-1} \quad \sigma_4(t_0) = -0.3 \text{ sec}^{-1}$$

Portray the time histories of

$$(i) \quad \underline{\omega}_1(t), \quad \underline{\vec{v}}_{p1}(t)$$

$$(ii) \quad \sigma_i(t) \quad (i=1, \dots, 4)$$

$$(iii) \quad \theta_i(t) \quad (i=1, \dots, 4)$$

Obtain the preceding plots for two cases

$$(1) \quad l = 2 \text{ m}, \quad \Delta t = 5 \text{ hrs}$$

$$(2) \quad l = 6 \text{ m}, \quad \Delta t = 30 \text{ min}$$

Comment on the time evolution of $\underline{\omega}_1$, σ_i , and θ_i .